Fiscal Year:	FY 2025	**Task Last Updated:**	FY 07/15/2026
PI Name:	Ocorr, Karen Ph.D.		
Project Title:	Integrated Physiological Responses of CNS and Muscle in Drosophila and C. elegans Along a Gravity Continuum		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Invertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology (2) Neurobiology		
Space Biology Special Category:	None		
PI Email:	kocorr@sbpdiscovery.org	**Fax:**	FY
PI Organization Type:	NON-PROFIT	**Phone:**	858-692-0051
Organization Name:	Sanford Burnham Prebys Medical Discovery Institute		
PI Address 1:	Center for Genetic Disorders and Aging Research		
PI Address 2:	10901 N Torrey Pines Rd		
PI Web Page:			
City:	La Jolla	**State:**	CA
Zip Code:	92037-1005	**Congressional District:**	49
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2020 Space Biology NNH20ZDA001N-SB E.12. Flight/Ground Research
Start Date:	01/01/2022	**End Date:**	06/30/2026
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:	Jones, Harry	**Contact Phone:**	650.604.5518
Contact Email:	harry.jones@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 03/60/2026 per F. Hernandez/NASA-ARC (Ed., 3/31/26). NOTE: End date changed to 12/31/2025 per F. Hernandez/NASA-ARC (Ed., 2/12/25).		
Key Personnel Changes/Previous PI:	Sylvain Costes, Ph.D. has retired from NASA. Nathaniel Szewczyk, Ph.D. has passed away. They are no longer on the list of Co-Investigators (Ed., 7/15/26).		
COI Name (Institution):	Iyer, Janani Ph.D. (NASA Ames Research Center) Mhatre, Siddhita Ph.D. (NASA Ames Research Center)		
Grant/Contract No.:	80NSSC22K0278		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	2026 UPDATE: The Principal Investigator (PI) reports that, following cancellation of the original mission, the team applied to continue the basic experiment of examining the effects of altered gravity using ground-based samples. The team received verification that it could pursue this altered gravity approach, and refocused the project as, "Integrated physiological and cellular responses of CNS and muscle in Drosophila under hyper g" (Ed., 7/15/26). ORIGINAL ABSTRACT: Our studies will use the fruit fly Drosophila and the worm Caenorhabditis elegans (C. elegans) to identify conserved mechanisms underlying the oxidative stress response to altered gravity. Spaceflight induces alterations in somatic/cardiac muscle, as well as in the brain. Many of these changes mirror those induced by long-term bed-rest on Earth and with age. We will use functional, structural, and molecular biological techniques to identify common genetic and molecular components that mediate the effect of microgravity, lunar gravity, and Mars gravity on organ function. The use of two different genetic model organisms will allow us to identify common targets across species that can be exploited to mitigate negative health effects of long duration space habitation and perhaps provide therapies to combat muscle wasting and neurodegeneration on Earth. We will also compare the changes in these organisms with published changes in humans subjected to bed-rest and spaceflight.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Understanding the effects of reduced gravities on mitochondrial structure/function and protein homeostasis will provide insights into the cellular mechanisms underlying sarcopenia and muscle wasting. Thus providing targets for counter measures and clinically relevant therapies.
Task Progress:	Our experiment was originally expected to fly late 2023. But Redwire, who handles the equipment we will use, only signed a contract in Oct. 2023. We spent 2024 testing that equipment and performing Science Verification Tests. We were tentatively scheduled to fly in early 2024. However, that flight was moved several times and the experiment was recently canceled. In addition, my Co-I, Nathaniel Szewczyk, unexpectedly died. Since then we have been mostly waiting for approval of the change in focus. Since receiving that in mid-November 2025, we have been gearing back up with training technicians and testing the new protocols needed for the changes to the experiments. Due to all the delays and downtime we will need to extend the timeline to the end of June for the completion of these experiments. This does not include the time needed to write up the results. This No Cost extension was recently approved.
Bibliography Type:	Description: (Last Updated: 07/15/2026)
Articles in Peer-reviewed Journals	Scott A, Willis CRG, Muratani M, Higashitani A, Etheridge T, Szewczyk NJ, Deane CS. "Caenorhabditis elegans in microgravity: An omics perspective." iScience. 2023 Jul 21;26(7):107189. Review. http://dx.doi.org/10.1016/j.isci.2023.107189 ; PMID: 37456835; PMCID: PMC10344948 , Jul-2023
Articles in Peer-reviewed Journals	Tasoula A, Szewczyk N. "Astronaut proteomics: Japan leads the way for transformative studies in space." Proteomics. 2024 Oct 10;24(20):e2300645. https://doi.org/10.1002/pmic.202300645Digital Object Identifier (DOI) ; PMID: 39388383 , Oct-2024
Articles in Peer-reviewed Journals	Rutter LA, MacKay MJ, Cope H, Szewczyk NJ, Kim J, Overbey E, Tierney BT, Muratani M, Lamm B, Bezdan D, Paul AM, Schmidt MA, Church GM, Giacomello S, Mason CE. "Protective alleles and precision healthcare in crewed spaceflight." Nat Commun. 2024 Jul 22;15:6158. https://doi.org/10.1038/s41467-024-49423-6 ; PMID: 39039045; PMCID: PMC11263583 , Jul-2024
Articles in Peer-reviewed Journals	Seylani A, Galsinh AS, Tasoula A, I AR, Camera A, Calleja-Agius J, Borg J, Goel C, Kim J, Clark KB, Das S, Arif S, Boerrigter M, Coffey C, Szewczyk N, Mason CE, Manoli M, Karouia F, Schwertz H, Beheshti A, Tulodziecki D. "Ethical considerations for the age of non-governmental space exploration." Nat Commun. 2024 Jun 11;15:4774. Review. https://doi.org/10.1038/s41467-023-44357-x ; PMID: 38862473; PMCID: PMC11166968 , Jun-2024
Articles in Peer-reviewed Journals	Rutter LA, Cope H, MacKay MJ, Herranz R, Das S, Ponomarev SA, Costes SV, Paul AM, Barker R, Taylor DM, Bezdan D, Szewczyk NJ, Muratani M, Mason CE, Giacomello S. "Astronaut omics and the impact of space on the human body at scale." Nat Commun. 2024 Jun 11;15:4952. Review. https://doi.org/10.1038/s41467-024-47237-0 ; PMID: 38862505; PMCID: PMC11166943 , Jun-2024
Articles in Peer-reviewed Journals	Camera A, Tabetah M, Castañeda V, Kim J, Galsinh AS, Haro-Vinueza A, Salinas I, Seylani A, Arif S, Das S, Mori MA, Carano A, de Oliveira LC, Muratani M, Barker R, Zaksas V, Goel C, Dimokidis E, Taylor DM, Jeong J, Overbey E, Meydan C, Porterfield DM, Díaz JE, Caicedo A, Schisler JC, Laiakis EC, Mason CE, Kim MS, Karouia F, Szewczyk NJ, Beheshti A. "Aging and putative frailty biomarkers are altered by spaceflight." Sci Rep. 2024 Jun 11;14:13098. https://doi.org/10.1038/s41598-024-57948-5 ; PMID: 38862573; PMCID: PMC11166946 , Jun-2024
Articles in Peer-reviewed Journals	Cope H, Elsborg J, Demharter S, McDonald JT, Wernecke C, Parthasarathy H, Unadkat H, Chatrathi M, Claudio J, Reinsch S, Avci P, Zwart SR, Smith SM, Heer M, Muratani M, Meydan C, Overbey E, Kim J, Chin CR, Park J, Schisler JC, Mason CE, Szewczyk NJ, Willis CRG, Salam A, Beheshti A. "Transcriptomics analysis reveals molecular alterations underpinning spaceflight dermatology." Commun Med (Lond). 2024 Jun 11;4(1):106. https://doi.org/10.1038/s43856-024-00532-9 ; PMID: 38862781; PMCID: PMC11166967 , Jun-2024
Articles in Peer-reviewed Journals	Mathyk BA, Tabetah M, Karim R, Zaksas V, Kim J, Anu RI, Muratani M, Tasoula A, Singh RS, Chen YK, Overbey E, Park J, Cope H, Fazelinia H, Povero D, Borg J, Klotz RV, Yu M, Young SL, Mason CE, Szewczyk N, St Clair RM, Karouia F, Beheshti A. "Spaceflight induces changes in gene expression profiles linked to insulin and estrogen." Commun Biol. 2024 Jun 11;7:692. https://doi.org/10.1038/s42003-023-05213-2 ; PMID: 38862620; PMCID: PMC1116698 , Jun-2024